

---

# FINMEM: A PERFORMANCE-ENHANCED LLM TRADING AGENT WITH LAYERED MEMORY AND CHARACTER DESIGN

---

Yangyang Yu\*, Haohang Li\*, Zhi Chen\*, Yuechen Jiang\*, Yang Li\*  
Denghui Zhang, Rong Liu, Jordan W. Suchow, Khaldoun Khashanah†

*Stevens Institute of Technology*  
Hoboken, NJ, United States

{yyu44, hli113, zchen100, yjiang52, yli269, dzhang42, rliu20, jws}@stevens.edu

## ABSTRACT

Recent advancements in Large Language Models (LLMs) have exhibited notable efficacy in question-answering (QA) tasks across diverse domains. Their prowess in integrating extensive web knowledge has fueled interest in developing LLM-based autonomous agents. While LLMs are efficient in decoding human instructions and deriving solutions by holistically processing historical inputs, transitioning to purpose-driven agents requires a supplementary rational architecture to process multi-source information, establish reasoning chains, and prioritize critical tasks. Addressing this, we introduce FINMEM, a novel LLM-based agent framework devised for financial decision-making. It encompasses three core modules: Profiling, to customize the agent’s characteristics; Memory, with layered message processing, to aid the agent in assimilating hierarchical financial data; and Decision-making, to convert insights gained from memories into investment decisions. Notably, FINMEM’s memory module aligns closely with the cognitive structure of human traders, offering robust interpretability and real-time tuning. Its adjustable cognitive span allows for the retention of critical information beyond human perceptual limits, thereby enhancing trading outcomes. This framework enables the agent to self-evolve its professional knowledge, react agilely to new investment cues, and continuously refine trading decisions in the volatile financial environment. We first compare FINMEM with various algorithmic agents on a scalable real-world financial dataset, underscoring its leading trading performance in stocks. We then fine-tuned the agent’s perceptual span and character setting to achieve a significantly enhanced trading performance. Collectively, FINMEM presents a cutting-edge LLM agent framework for automated trading, boosting cumulative investment returns.

**KEYWORDS:** Financial AI, Large Language Model, Trading Algorithms, Deep Learning, Financial Technology

## 1 Introduction

With the influx of diverse financial data streams from the web, traders face a deluge of information from various sources. This requires them rapidly to understand, memorize, and filtrate crucial events for investment decisions. However, innate cognitive limitations restrict human traders from processing information within their perception and memory capacity, a span much narrower than the actual volume of available information [5]. Consequently, insufficiently considering or even dismissing critical events affecting trading decisions becomes increasingly concerning as data availability expands. To overcome the physical limitations in the memory systems of human traders, researchers have been consistently working on designing autonomous trading agent systems. These systems need to thoroughly integrate all available information and possess a sophisticated design in the agent’s backbone algorithm to deliver enhanced trading performance.

The evolution of autonomous trading systems has transitioned from the initial rule-based trading strategies [12] to more advanced machine-learning-based algorithms [19]. In recent years, Reinforcement Learning (RL)-based agents [14],

†Corresponding author. Email: khashan@stevens.edu

\*Equal contribution, with author order decided by dice roll.

<sup>2</sup>The source code of this project can be found via: FINMEM LLM Trading

especially those employing Deep Reinforcement Learning (DRL) [29] as backbone algorithms, garner joint attention of both academia and industry. Leveraging both RL principles and deep learning, DRL agents effectively handle and learn from scalable and diverse financial data, including stock prices, key financial indicators, and market sentiments. They utilize deep neural networks to extract expressive features from input data, representing the complex financial market environment, which enhances their comprehension ability. Retaining the key features of RL agents, they learn through interaction with a predefined environment to maximize investment gain over time. Research suggests DRLs can meet the crucial needs of trading agents to process and make informed decisions from large volumes of data. However, certain inherent features of DRL algorithms exhibit notable deficiencies in financial applications. **Firstly, DRL agents exhibit a lack of interpretability concerning the rationale behind their decisions [4].** They are often described as “black boxes,” where the internal processes and computational layers leading to a specific decision are neither easily understandable nor transparent. **Secondly, DRL agents find it challenging to effectively integrate textual data with numerical features.** Text data plays a vital role in finance, since the majority of market information is conveyed through news articles and financial reports. However, transforming text data to embeddings considerably increases input space dimensionality, making learning more computationally demanding. Plus, empirical studies have shown that combining textual representations with numerical financial indicators often leads to convergence challenges [15]. Therefore, a backbone algorithm with transparent reasoning and the enhanced ability to capture investment-related textual insights comprehensively is essential.

Recent advancements in Large Language Models (LLMs), like Generative Pre-trained Transformers (GPTs) [34], offer viable solutions for developing trading agents, alleviating previous concerns. With carefully designed prompts, the LLM-based agent is able to provide reasons and outcomes in plain text. This allows for immediate observation and prompt adjustment of its reasoning process. Employing LLMs as the backbone algorithms for agents also overcomes the constraint of isolated environments. This is achieved through their vast pre-existing knowledge and the effective integration of valuable insights from a variety of data sources, including both textual and numerical. When equipped with suitable prompt templates, this approach significantly enhances decision-making capabilities [50]. Studies indicate that prompt-guided reasoning significantly improves problem-solving rates across various domains [24]. Notably, a growing body of research has focused on utilizing LLMs to make informed trading decisions for stocks and funds by continuously interacting with financial environment information [55, 52]. However, in currently available approaches, LLMs primarily serve as a QA role rather than functioning as autonomous agents. **The potential issue with these approaches is the incapability of fully understanding the varying timeliness associated with different types of financial data.** These financial LLM agents, despite outperforming traditional trading benchmarks, generally process information indiscriminately through QA iterations, lacking the ability to memorize influential messages. Furthermore, their method of acknowledging the timeliness of financial data is heavily dependent on the uncertain and laborious LLM fine-tuning process. These insufficiencies undermine their ability to update the knowledge base in a daily manner, meaning they lack a memory component. As a result, they may struggle to prioritize significant and influential memory events effectively. Additionally, current literature on LLM-based trading agents lacks a comparative analysis between these applications and other autonomous trading systems, such as DRL agents.

To bridge this gap, we present FINMEM, an innovative LLM-based autonomous trading agent with a novel layered memory system and dynamic character design. Unlike previous LLM agents in finance, FINMEM encompasses a memory module adept at processing multi-source financial data with varying timeliness and self-adaptive character setting for fitting into volatile market environments.

Our concept is initially inspired by the Generative Agents framework by Park et al. [35], aimed at enhancing the efficient retrieval of key events for general-purpose LLM agents. This framework features a unique character design and seed memory, activating the agent upon specific query through prompts. It prioritizes events in a unified memory stream, ranked by a linear combination of recency, relevancy, and importance. The framework outlined in [35] provides a foundational structure for LLM agent design. It includes a profiling module for character definition, a memory module for experience recording and critical information retrieval, and an action module to guide actions based on the retrieved memories. This structure effectively facilitates goal achievement for the agent in a general social environment. **However, Park et al.’s framework struggles with comprehending financial data with varying timeliness and importance,** like daily news versus quarterly and annual reports. Key challenges involve quantifying the timeliness of different information sources, optimizing information retrieval, and enhancing trading decisions with detailed analysis. To tackle these challenges, we further propose FINMEM with the following improvements.

FINMEM maintains a modular approach similar to Park et al. [35], but features novel design of profiling and memory modules. Its specialized profiling module equips FINMEM with a trading-task-specific professional background, enhancing robustness to market fluctuations via offering self-adaptive risk inclination option. FINMEM’s memory module innovatively incorporates working memory and layered long-term memory components, ideal for stratified information processing. Its working memory acts as a dynamic “workspace,” enabling operations like summarization, observation, and reflection on multi-source information to facilitate trading decisions. Its long-term memory, structured

into shallow, intermediate, and deep layers [9], manages varied decay rates to satisfy the need to retain distinct types of financial information within different time scales based on their corresponding timeliness. For instance, daily news, with its immediate effects on stock markets, is channeled into the shallow processing layer. Meanwhile, annual company reports, exerting a more prolonged impact, are processed in the deep layer by FINMEM. Each layer in FINMEM prioritizes memory events based on the assemble of recency, relevancy, and importance close to Park et al.’s method. However, it introduces new measurements for recency and importance, specifically tailored to better rank financial data according to their unique time sensitivity. FINMEM’s memory mechanism can also transit significantly impactful investment memory events to deeper processing layers, ensuring their retention for extended periods. FINMEM’s memory module can mirror the human cognitive system [47] and facilitate agile, real-time decisions [46]. It enables continuous evolution in professional knowledge through structured summarizing, retrospecting past experiences, and reacting to new trading scenarios. Additionally, FINMEM includes a decision-making module capable of deriving investment decisions by considering top-ranked memory events and current market conditions.

Through experiments, we show that FINMEM exhibits an outstanding ability to stratify and leverage the various levels of market insights, significantly improving the quality of trading decisions. We claim that FINMEM provides these key contributions:

**FINMEM presents a state-of-the-art LLM-based trading agent with a human-aligned memory mechanism and character design**, particularly crafted to capture investment insights from the financial market. In its agent memory module design, FINMEM innovatively emulates human working and layered long-term memory mechanisms. This approach effectively harnesses the time-sensitive aspects of financial data, capturing crucial investment insights and thereby boosting trading performance. FINMEM’s profiling module includes a dynamic character setting feature, offering seed information about professional backgrounds and adjustable risk inclinations. Additionally, it continuously updates domain knowledge as the trading experience grows, thereby enhancing FINMEM’s capabilities. Ablation studies demonstrate that FINMEM can learn from past trading experiences and evolve its knowledge base through continuous market interaction, maintaining robustness in complex markets for profitable trading decisions.

**FINMEM can utilize its distinctive features to expand the agent’s perceptual range beyond the human limitation to make well-informed trading decisions.** Cognitive research indicates that human working memory can recall only five to nine events at a time [30]. This limitation, while preventing information overload, can yield insufficient insight for precise decision-making. In contrast, FINMEM’s memory module transcends this constraint. It allows adjusting cognitive load by selecting a flexible number of top-ranked events from each layer of its hierarchical long-term memory, allowing FINMEM to maintain agility and deliver superior trading decisions in data-rich contexts.

**FINMEM achieves impressive trading performance using training data that is limited in volume and spans a short time period.** Our experiments indicate that training FINMEM with daily collected data over just six months to a year is enough to produce robust and notable trading results. This timeframe is considerably shorter than what other comparable models require. This efficiency is achieved through the optimal utilization of multi-source incoming data and the precise identification of key signals for trading decisions. Furthermore, it’s noteworthy that FINMEM’s effectiveness is demonstrated on smaller datasets and with general-purpose LLMs. Its capabilities are anticipated to be further amplified with access to larger, higher-quality financial datasets and LLMs fine-tuned specifically for financial applications.

In this paper, we begin by explaining the three core modules of FINMEM. Subsequently, we emphasize its superior trading performance compared to a range of representative algorithmic agents. We further explore how FINMEM achieves its optimal performance, examining adjustments in three key aspects: backbone algorithms, working memory capacity, and character settings.

## 2 Related Work

### 2.1 Backbone Algorithms of Contemporary Autonomous Trading Agents

The development of trading agents has evolved over several decades, influenced by advancements in technology, finance, and computational methodologies. Conventionally, a rule-based algorithm for trading stocks is an automated strategy that operates based on a predefined set of rules[7, 49, 37]. These rules are often derived from historical market patterns and trading experience. Compared with rule-based algorithms that use predefined rules and conditions, Reinforcement Learning provides a way for agents to learn by interacting with an environment and receiving feedback in the form of rewards or penalties. [10, 22]. Deep learning models can be integrated with RL to handle large and complex state spaces, like those in stock markets. Such models are often referred to as Deep Reinforcement Learning (DRL) [53, 54]. For example, Deep Q-Network (DQN) [44], Advantage Actor-Critic (A2C) [56], and Proximal Policy Optimization (PPO) [27] are popular algorithms for such tasks. Using DRL agents as automated financial trading backbones, face